A Möbius-binomial C/F bridge and a claim-disciplined theorem-transfer atlas for Apéry-like sources

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Abstract

We isolate a direct Möbius-binomial bridge between Zagier's C and F Apéry-like sequences:

$$F(x) = (1-9x)^{-1}C\left(\frac{-x}{1-9x}\right).$$

Equivalently, the bridge is an involutive binomial coefficient transform between the two catalog-normalized coefficient sequences; it also transports the differential equation of C to that of F and is cross-checked by the known Franel transform route. We then use this bridge as a primitive for a theorem-transfer atlas. The atlas distinguishes closed internal results, theorem imports by reference, conditional transfers, finite exact replays, source-attached conjectures, proof obligations, blocked rows, bounded negatives, watch metadata, and closed metadata. In particular, we do not derive Sun's Conjecture 3.5 from the C/F bridge, do not promote finite replays or coefficient matches to theorems, and do not turn Dwork or Rowland–Yassawi metadata into unconditional theorem claims. Sun's raw-C conjecture is handled by a separate Franel–Domb bridge proof [Bab26]. The result here is a claim-disciplined package for using a proved C/F bridge while keeping adjacent Apéry-like source layers separate.

1 Introduction

Apéry-like sequences sit at the intersection of several structures: recurrences, modular parametrizations, binomial transforms, constant-term and Laurent-polynomial models, supercongruences, and automatic congruences. These structures often appear in clusters, but they do not transfer automatically from one sequence to another. A coefficient match, a shared recurrence shape, or a common modular level may suggest a relation; none of these is by itself a theorem-transfer mechanism.

This point of view is close in spirit to the tabular literature around Calabi–Yau and Apéry-like differential equations, where large lists of operators, transformations, and periods are useful only when their normalizations and equivalences are explicit [AvEvSZ05, AvSZ11]. Our scope is much narrower: a claim-disciplined atlas centered on one proved C/F bridge.

This paper is organized around one case where the transfer mechanism is explicit. We prove and certify a direct Möbius-binomial bridge between the Zagier C and F sequences:

$$F(x) = (1-9x)^{-1}C\left(\frac{-x}{1-9x}\right).$$

The identity gives an involutive coefficient transform, an ODE pullback, and a constant-term complement operation. It is compatible with the known Franel network around these sequences.

The second contribution is methodological. We use the C/F bridge as a primitive in a theorem-transfer atlas for nearby Apéry-like source layers. The atlas records what can be transferred, what can only be imported by reference, what is merely metadata, and what remains a proof obligation. The point is not to claim that all adjacent congruence frameworks become theorems of this paper. Rather, we separate proof, imported theorem, finite replay, conjectural source attachment, blocked metadata, watch rows, and bounded negative evidence.

The bridge is the mathematical core; the atlas records how far its transport consequences extend. Sections 2–4 give the C/F bridge and its certificate. Sections 5–11 explain how the bridge is used to build a disciplined theorem-transfer atlas. Appendices contain the theta-gauge certificate, the constant-term complement lemma, selected registry rows, open problems, and source-status notes.

2 Zagier C and F

Let C_n and F_n denote Zagier's C and F Apéry-like sequences in the catalog convention used throughout this paper [Zag09, OEI26a, OEI26b]. Write

$$C(x) = \sum_{n\geq 0} C_n x^n, \qquad F(x) = \sum_{n\geq 0} F_n x^n.$$

The C sequence is given by

$$C_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{2k}{k}.$$

It satisfies

$$(n+1)^2 C_{n+1} = (10n^2 + 10n + 3)C_n - 9n^2 C_{n-1}.$$

The F sequence satisfies

$$(n+1)^2 F_{n+1} = (17n^2 + 17n + 6)F_n - 72n^2 F_{n-1}.$$

The atlas below always treats these catalog-normalized objects as the named source and target rows.

3 The C/F Möbius-binomial bridge

3.1 Modular parameter and theta-gauge normalization

We use the modular parametrizations recorded by Beukers in the common $\Gamma_0(6)$ setting [Beu25]. In the C row, let $t_C(q)$ denote the source modular parameter and put

$$\Theta_C(q) = C(t_C(q)).$$

For the F row, Beukers' source parameter is twist-guarded. We write it as $t_F^{\mathrm{source}}(q)$ and use the catalog parameter

$$t_F(q) = t_F^{\mathrm{cat}}(q) := -t_F^{\mathrm{source}}(q).$$

The F theta series used below is therefore

$$\Theta_F(q) = F(t_F(q)) = F(-t_F^{\mathrm{source}}(q)) = \Theta_F^{\mathrm{source}}(q).$$

All q-expansions, cusp-order checks, and Sturm checks in this paper are in this common normalization. In particular, the two identities used in the bridge proof are identities of meromorphic modular functions on $X_0(6)$ in this normalization.

Theorem 3.1 (Zagier C/F Möbius-binomial bridge)**.** *Under the catalog convention used here and the verified F twist transport,*

$$F(x) = (1-9x)^{-1} C\left(\frac{-x}{1-9x}\right).$$

Proof. With the normalization of Section 3.1, the theta-gauge identities are

$$\Theta_F/\Theta_C = 1 - 9t_C, \qquad \Theta_C/\Theta_F = 1 + 9t_F.$$

With

$$u = t_C(q), \qquad x = t_F^{\mathrm{cat}}(q) = \frac{-u}{1-9u},$$

one has

$$\frac{-x}{1-9x} = u, \qquad (1-9x)^{-1} = 1 - 9u.$$

Thus

$$F(x) = \Theta_F(q) = (1-9u)\Theta_C(q) = (1-9x)^{-1}C\left(\frac{-x}{1-9x}\right).$$

The theta-gauge identities are verified by forward and inverse valence certificates, with a cleared-denominator Sturm backup; see Appendix A. ☐

Corollary 3.2 (Coefficient transform)**.** *Define*

$$T(a)_m = \sum_{n=0}^{m} \binom{m}{n} (-1)^n 9^{m-n} a_n.$$

Then $T(C) = F$ and $T(F) = C$.

Proof. Taking coefficients in Theorem 3.1 gives $T(C) = F$. The substitution $x \mapsto -x/(1-9x)$ is involutive with the same prefactor contract, giving $T(F) = C$. ☐

Corollary 3.3 (ODE pullback)**.** *Let*

$$L_C = x(x-1)(9x-1)D_x^2 + (27x^2 - 20x + 1)D_x + 3(3x-1)$$

be the differential operator associated to the C recurrence, and let

$$L_F = x(72x^2 - 17x + 1)D_x^2 + (216x^2 - 34x + 1)D_x + 6(12x-1)$$

be the corresponding F operator. Under

$$u = -\frac{x}{1-9x}, \qquad F(x) = (1-9x)^{-1}C(u),$$

symbolic substitution gives

$$L_C[C](u) = (9x-1)L_F[F](x).$$

Thus the C equation pulls back to the F equation up to the nonzero rational prefactor $9x - 1$.

Remark 3.4 (Franel route)**.** *Let $R_n = \sum_{k=0}^{n} \binom{n}{k}^3$. The known Franel transforms are*

$$C(x) = (1-x)^{-1}R\left(\frac{x}{1-x}\right), \qquad F(x) = (1-8x)^{-1}R\left(\frac{-x}{1-8x}\right),$$

with provenance through Barrucand, Callan, Verrill, and OEIS [Bar75, Cal08, Ver10]. Composing these formulas gives the same C/F bridge.

4 Certificate and normalization

The proof depends on four normalization checks. First, C and F are placed in a common $\Gamma_0(6)$ normalization. Second, the source F parameter is transported to the catalog-normalized F twist. Third, the forward and inverse theta-gauge identities are certified by cusp-order and valence arguments. Fourth, a cleared-denominator Sturm check gives an independent backup. The main text uses only the resulting identities; the certificate details are recorded in Appendix A.

5 Transport contracts generated by the bridge

5.1 Coefficient transform as transport

The coefficient transform in Corollary 3.2 is the atlas primitive. It allows a theorem stated for F to be rewritten as a theorem for $T(C)$. It does not automatically produce a theorem for raw C_n.

For example, Sun's source rows include congruences for a sequence Q_n with $Q_n = (-1)^n F_n$ in the relevant convention [Sun22]. The bridge rewrites these as statements about $(-1)^n T(C)_n$. Sun's raw-C conjectural congruence is not a consequence of this C/F bridge; it is resolved separately by a Franel–Domb bridge through intermediate Franel, Domb, and Rodriguez–Villegas sums [Bab26].

5.2 Franel network

Remark 3.4 explains why the bridge is not an isolated accident: C and F are both connected to the Franel sequence by known Möbius-binomial transforms. The theorem-transfer atlas records this as known source metadata. It does not treat every inverse or coefficient replay as a new theorem edge.

More concretely, both generating functions are obtained from the same cubic Franel series $R(x) = \sum R_n x^n$ by two different fractional-linear changes of variable. The C/F bridge is therefore the composition of two classical Franel transforms, not a relation inferred only from matching initial coefficients. This Franel viewpoint is useful in the atlas because it separates a known structural network from later theorem-transfer claims.

5.3 Constant-term complement

The coefficient transform has a direct constant-term interpretation.

Lemma 5.1 (Binomial-complement constant-term transform)**.** *Let Λ be a Laurent polynomial and let*

$$A_n = \mathrm{CT}(\Lambda^n).$$

For a scalar a, define

$$B_n = \sum_{k=0}^{n} \binom{n}{k} (-1)^k a^{n-k} A_k.$$

Then

$$B_n = \mathrm{CT}((a - \Lambda)^n).$$

Proof. By the binomial theorem,

$$(a - \Lambda)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} (-\Lambda)^k.$$

Taking constant terms gives

$$\mathrm{CT}((a - \Lambda)^n) = \sum_{k=0}^{n} \binom{n}{k} (-1)^k a^{n-k}\, \mathrm{CT}(\Lambda^k) = B_n.$$

□

For

$$\Lambda_C = \frac{(x + y + 1)(xy + x + y)}{xy}$$

Gorodetsky records this Laurent model for the C row [Gor23]. Equivalently, if $P = (x + y + 1)(xy + x + y)$, then

$$\mathrm{CT}(\Lambda_C^n) = [x^n y^n]P^n,$$

and the cited model identifies this coefficient with the catalog-normalized C_n. Lemma 5.1 gives

$$F_n = \mathrm{CT}((9 - \Lambda_C)^n).$$

This is a constant-term operation. It is not a Laurent mutation claim and it is not a Newton-equivalence claim between $9 - \Lambda_C$ and other F-side models such as the Gorodetsky replacement model [Gor23].

5.4 Rational diagonals and small automata

The constant-term models also admit rational diagonal contracts, which support small automatic-congruence metadata for C/F and related rows. This paper records those automata as metadata. It does not claim a full Rowland–Yassawi theorem-by-reference application, because that would require the source-level bad-prime and singularity filters in addition to an explicit Cartier-state construction [RY15].

6 Claim classes and atlas discipline

The atlas uses explicit claim classes. Table 1 summarizes the ones used in this paper.

`closed_internal_result`	Proved or certified inside the C/F package.
`theorem_import_by_reference`	External theorem used with its source label; not renamed as an internal theorem.
`conditional_theorem_transfer`	Transfer valid after an explicit bridge or source assumption.
`closed_metadata`	Verified auxiliary record, such as a normalization map, fixed-modulus automaton, source-label reconciliation, or known literature identity; not used as a new theorem unless separately promoted through a proof route.
`explicit_automaton_metadata`	Explicit finite-state or prime-power automaton data recorded as metadata; in the V2 package this is treated as a closed-metadata subtype, not as a theorem-transfer class.
`finite_exact_replay`	Exact finite check used as evidence only.
`source_attached_conjecture`	Conjectural source row attached to the atlas without proof.
`proof_obligation`	A precise missing lemma, not a vague direction.
`blocked`	A known missing contract prevents theorem use.
`watch`	Interesting metadata without a promotion route.
`bounded_negative`	A bounded search failed a stated gate.

Table 1: Claim classes used in the theorem-transfer atlas.

The guardrails are simple: finite replay is evidence, not proof; coefficient matching is not theorem transfer; and external theorems remain external theorems.

7 The theorem-transfer registry

The registry is an operational table of theorem imports and bridge-generated transfers. The complete table is part of the data package; Appendix C records the schema and selected rows. The core examples are:

1. the C/F bridge, a closed internal result;
2. the coefficient involution, a closed internal result;
3. the ODE pullback, a closed internal result;
4. Sun's F-side source rows transferred only to $T(C)$;
5. Sun's raw-C target, retained in the historical V1/V2 atlas as a source-attached conjecture and proof obligation, but now marked in the paper-facing readout as resolved separately by [Bab26];
6. CT/Newton guardrails, which block mutation or Newton-equivalence claims without an explicit operation.

8 Three source-layer examples

8.1 Sun layer

Sun supplies source congruences for F-side objects and related binomial sums [Sun22]. The C/F bridge transfers F-side statements to $T(C)$. It does not prove raw-C variants from C/F data. The raw-C Conjecture 3.5 is a separate Franel–Domb theorem [Bab26], and the atlas keeps that proof route separate from the C/F theorem-transfer rows.

8.2 Dwork layer

Mellit–Vlasenko give Dwork-type congruences for constant terms of powers of Laurent polynomials [MV16]. The complement model $9 - \Lambda_C$ gives an additional F-side constant-term witness. We record it as Dwork metadata by reference. We do not claim a Newton equivalence or a Laurent mutation between this complement model and other F-side models.

8.3 Automatic congruence layer

Rowland–Yassawi provide the source framework for automatic congruences of rational diagonals [RY15]. The atlas records explicit small automata for Cooper, Franel, and C/F small moduli, including C/F mod 2, mod 3, mod 4, and mod 9 metadata. These are explicit automaton or prime-power metadata; they are not a full Rowland–Yassawi theorem import.

Other source layers, including Liu's binomial-transform congruence profile, Straub's Gessel–Lucas theorem, and Beukers–Tsai–Ye's modular-form Lucas congruence source [Liu25, Str24, BTY25], are summarized in Appendix E. They are kept brief here because their role is source attachment and claim discipline, not the main C/F theorem.

9 Open-problem atlas

The open-problem atlas records blockers rather than vague directions. A selection is shown in Table 2; a fuller schema appears in Appendix D.

10 Edge scan and watch-triage discipline

A bounded C/F-like Möbius-binomial scan checked 17684 transforms of the form

$$B(x) = (1 - ax)^{-r} A\left(\frac{sx}{1 - ax}\right).$$

Branch	Blocking lemma or missing contract	Status
BTY weight-one rows	row-specific $\chi(w_d)$ values	open
Straub derivative transfer	low-carry identity for the corrected connection	proof obligation
Dwork model comparison	controlled equivalence between complement model and Q_F	open
Rowland–Yassawi source application	bad-prime/singularity filter and source-style Cartier states	inactive
Edge-watch scan	no unresolved V1 watch rows remain	closed metadata
Sun raw-C to RV-108	Franel–Domb raw-C/RV bridge outside the C/F atlas	resolved separately
A2b global glue	Domb branch-multiplier identity outside the C/F atlas	resolved separately

Table 2: Selected open-problem and blocker rows.

It recovered known or canonical relations and promoted no new theorem edge. The eight previous watch hits were subsequently resolved as known metadata: six are alias/canonical identity hits for the Apéry zeta(3) and Almkvist–Zudilin gamma catalog rows, and two are inverse Franel-transform hits. This is a negative discipline result: coefficient matches are not automatically promoted.

11 Data package and reproducibility

The data package contains registry rows, claim classes, source labels, proof obligations, open-problem statuses, edge-scan triage records, manifests, and readouts. The paper-facing purpose is reproducibility of claim status: a reader should be able to tell which rows are proved internally, which are imported by reference, which are finite replays, and which are blocked. The package is based on the V2 manifest, with the V1 registry retained as immutable provenance for the original C/F theorem-transfer rows. Appendix E lists source identifiers and package fields; the main text avoids path dependencies. The final public version will attach this curated, repo-relative data package as supplementary material or through a public repository before submission.

12 Conclusion

We do not solve every problem attached to the C/F bridge. Instead, we give a normalized proof of the bridge and a reproducible model for using it without overclaiming. The bridge transports generating functions, coefficients, ODEs, and a constant-term complement model. The atlas records adjacent theorem sources, metadata layers, finite checks, conjectures, blockers, and bounded negatives with distinct claim classes. This separation is the main lesson: bridge identities are powerful precisely when their scope is explicit.

A Theta-gauge certificate details

This appendix records the certificate data behind the theta-gauge identities used in Theorem 3.1.

By the normalization contracts in Section 3.1, the functions below are meromorphic modular functions on $X_0(6)$ and have no interior poles. Their only possible poles are cuspidal; the cusp-order tables bound those poles.

A.1 Forward valence certificate

The forward identity is

$$\Theta_F/\Theta_C = 1 - 9t_C.$$

Define

$$H(q) = \Theta_F(q)/\Theta_C(q) - (1 - 9t_C(q)).$$

The cusp-order audit on $\Gamma_0(6)$ gives

object	0	1/2	1/3	∞
t_C	0	-1	0	1
Θ_F/Θ_C	1	-1	0	0.

Thus the only possible pole of H is at the cusp 1/2, of order at most 1. The q-expansion replay gives $\mathrm{ord}_\infty(H) \geq 201$. Since this exceeds the possible pole order, the valence criterion forces $H = 0$.

A.2 Inverse valence certificate

The inverse identity is

$$\Theta_C/\Theta_F = 1 + 9t_F.$$

Define

$$K(q) = \Theta_C(q)/\Theta_F(q) - (1 + 9t_F(q)).$$

The inverse cusp-order audit gives

object	0	1/2	1/3	∞
t_F	-1	0	0	1
Θ_C/Θ_F	-1	1	0	0.

Thus the only possible pole of K is at the cusp 0, of order at most 1. The q-expansion replay gives $\mathrm{ord}_\infty(K) \geq 201$, and the valence criterion gives $K = 0$.

A.3 Sturm clear-denominator backup

The same two identities are also checked after clearing the possible cuspidal poles. Use the eta quotient

$$M(q) = \frac{\eta(q)^3\eta(2q)^3}{\eta(3q)\eta(6q)}.$$

By the Ligozat criterion, the exponent vector of M satisfies the two congruences modulo 24 and the total weight is 2. The cusp orders of M on $\Gamma_0(6)$ are

cusp	0	1/2	1/3	∞
$\mathrm{ord}(M)$	1	1	0	0.

Thus M is holomorphic at every cusp and vanishes at the two cusps where a pole can occur in the preceding valence argument. The deduction is explicit:

- H has no possible pole except at 1/2, of order at most 1, while $\mathrm{ord}_{1/2}(M) = 1$; hence MH is holomorphic at 1/2.
- At the other cusps H is holomorphic and M is holomorphic, so MH is holomorphic there as well.
- K has no possible pole except at 0, of order at most 1, while $\mathrm{ord}_0(M) = 1$; hence MK is holomorphic at 0.
- At the other cusps K is holomorphic and M is holomorphic, so MK is holomorphic there as well.

Since H and K have weight 0, both MH and MK are holomorphic weight-2 forms on $\Gamma_0(6)$, with the multiplier or character fixed by the same normalization. Their checked q-expansions lie in the rational coefficient field used by the certificate, so the ordinary Sturm criterion applies to the cleared forms. Since

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(6)] = 12, \qquad \left\lfloor \frac{2 \cdot 12}{12} \right\rfloor = 2,$$

the Sturm bound is 2. The replay gives vanishing at infinity far beyond this bound, hence both cleared forms vanish identically.

A.4 F twist transport dependency

The F-side parameter in source notation is twist-guarded. The transport contract is

$$F_{\mathrm{cat}}(-t_{\mathrm{source}}(q)) = \Theta_{\mathrm{source}}(q),$$

or equivalently

$$t_F^{\mathrm{cat}}(q) = -t_F^{\mathrm{source}}(q).$$

This is why Theorem 3.1 is stated in catalog-normalized F notation.

B Binomial-complement constant-term lemma

Lemma 5.1 is included in the main text because it is the cleanest bridge between coefficient transforms and constant-term models. The important guardrail is repeated here: the operation

$$\Lambda \mapsto a - \Lambda$$

preserves the fact that one has a constant-term sequence, but it is not a Laurent mutation claim and does not imply Newton equivalence between two chosen Laurent models.

C Registry schema and selected rows

Each registry row uses the following schema:

`statement_id`	row identifier
`source_paper`	external or internal source
`source_label`	the theorem, conjecture, or artifact label
`source_object`	source sequence or model
`target_object`	target sequence or model
`rewrite`	transport or normalization formula
`cf_role`	how the C/F bridge enters
`claim_level`	claim class
`finite_replay`	finite evidence, if any
`literature_status`	source status
`proof_obligation`	missing lemma, if any
`status`	current row status
`next_action`	next audit/proof step.

Selected rows include the C/F bridge, coefficient involution, ODE pullback, Sun F-side transfer to $T(C)$, the historical raw-C Sun proof-obligation row with a separate-paper resolution note, Dwork complement metadata, Liu convention metadata, and the edge-watch triage closeout.

D Open problems and blocking lemmas

The atlas records open work by the mathematical object that is missing. For the Sun branch, the C/F theorem transports F-side statements to $T(C)$, but it does not itself supply the raw-C/RV-108 bridge. In the historical V1/V2 data package this appears as a deferred proof obligation. After that atlas freeze, the missing raw-C/RV bridge and its A2b Domb component were supplied by the separate Franel–Domb bridge proof of Sun's Conjecture 3.5 [Bab26]. We keep the registry row as provenance for the C/F atlas rather than reclassifying it as a C/F consequence.

The modular-form and constant-term branches have different blockers. The Beukers–Tsai–Ye weight-one rows require row-specific multiplier data $\chi(w_d)$ before they can be used as theorem imports. The Dwork branch has a proved C/F constant-term complement operation, but the stronger comparison between the complement model and the Gorodetsky F-side model remains an equivalence-or-obstruction problem. The Rowland–Yassawi branch has explicit small automata as metadata, but a full source-theorem application would still require a bad-prime/singularity filter and source-style Cartier states.

Finally, the Straub derivative-transfer branch is not closed by the formal C/F coefficient involution alone. The corrected connection term leaves a finite low-carry identity as the remaining proof obligation. The edge-watch scan no longer contributes an open problem: its eight watch rows have been resolved as known metadata.

E Source and bibliography status

Zagier and OEIS fix the catalog-normalized C and F rows used throughout the paper. Beukers supplies the modular parametrizations, the common $\Gamma_0(6)$ setting, and the F twist caveat used in the theta-gauge certificate. Barrucand, Callan, and Verrill supply the classical Franel transform context that explains the C/F bridge as part of a known network rather than as an isolated coefficient coincidence.

The remaining sources enter as theorem imports or metadata layers. Sun supplies the source congruence layer and the original Conjecture 3.5 target [Sun22]; the latter is resolved by a separate Franel–Domb bridge proof rather than by the C/F bridge [Bab26]. Liu supplies binomial-transform congruence metadata with convention guards [Liu25]. In the Liu audit, thirteen rows match the normalized catalog directly, while the η and $s18$ rows are stored with an explicit convention map rather than as failures. Straub's Gessel–Lucas theorem is imported by reference where its hypotheses are used, but the separate C/F derivative-transfer problem remains guarded by its low-carry proof obligation [Str24].

The Beukers–Tsai–Ye theorem is used only when row hypotheses are complete [BTY25]. Mellit–Vlasenko and Gorodetsky provide the Dwork and constant-term context, and Rowland–Yassawi provides the source contract for automatic congruences [MV16, Gor23, RY15]. The broader tabular background of Calabi–Yau and Apéry-like operators is represented by the Almkvist–van Enckevort–van Straten–Zudilin tables and the subsequent transformation literature [AvEvSZ05, AvSZ11]. Before submission, the remaining bibliographic work is journal-specific style and final access-date policy for web catalog entries.

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